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IIT Madras

BS DEGREE

Candidate name Utkarsh Sahu

Roll No. DS21F1001107

Subject Linear Statistical Models

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Exam Session: Forenoon / Afternoon ☒

No. of sheets used 7

No. of pages used 14

Paper+Pen Exams – Instructions

(Please read these instructions carefully before starting the exam.)

1. Questions will be provided on the computer. Blue coloured page is the cover sheet where you have to fill your details. DO NOT write your answers in this Blue coloured sheet.
2. Yellow coloured pages with IIT Madras B.S. Degree header will be given to each candidate as Answer sheets. You have to mark/write on these Yellow -coloured answer sheets, tie the sheets together and return the same to the invigilator.
3. Answers written on any other paper will not be evaluated.
4. You have to write your Name & Roll Number in all Answer Sheets. Answer Sheets without your name and roll number will not be evaluated.
5. Tying the Answer Sheets securely is the candidate's responsibility.
6. Please DO NOT bring any type of Calculator for use in the exam. On-screen calculator will be available during the exam.
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- The End -



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Linear Statistical Models

(Q1)

Given,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

&

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$\hat{\beta}_0$ & $\hat{\beta}_1$ are unbiased estimator of β_0 & β_1 respectively if and only if:

$$* E(\hat{\beta}_1) = \beta_1 \quad \& \quad * E(\hat{\beta}_0) = \beta_0$$

Now,

$$E(\hat{\beta}_1) = E\left(\frac{\sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i + \epsilon_i)}{S_{xx}}\right)$$

$$\left[\text{assuming } S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \right]$$

$$= E\left(\beta_0 \frac{\sum_{i=1}^n (x_i - \bar{x})}{S_{xx}} + \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x}) x_i}{S_{xx}} + \frac{\sum_{i=1}^n (x_i - \bar{x}) \epsilon_i}{S_{xx}}\right)$$

$$= \beta_0 E\left(\frac{\sum (x_i - \bar{x})}{S_{xx}}\right) + \beta_1 E\left(\frac{\sum (x_i - \bar{x}) x_i}{S_{xx}}\right) + E\left(\frac{\sum (x_i - \bar{x}) \epsilon_i}{S_{xx}}\right)$$

Now we know,

$$\sum (x_i - \bar{x}) = 0$$

$$\sum (x_i - \bar{x})(x_i) = \sum (x_i - \bar{x})(x_i - \bar{x}) = S_{xx}$$

So, $E(\epsilon_i) = 0 \Rightarrow E(\sum (x_i - \bar{x}) \epsilon_i) = 0$

$$E(\hat{\beta}_1) = \beta_0(0) + \beta_1(1) + 0$$

$$= \beta_1$$

Since, $E(\hat{\beta}_1) = \beta_1$ Hence $\hat{\beta}_1$ is unbiased estimator.

Now, for $\hat{\beta}_0$

Given,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\Rightarrow E(\hat{\beta}_0) = E(\bar{y}) - E(\hat{\beta}_1 \bar{x})$$

$$= E\left(\frac{\sum y_i}{n}\right) - \bar{x} E(\hat{\beta}_1) \quad [\bar{x} \text{ is constant}]$$

$$= \frac{1}{n} \sum E(y_i) - \bar{x} E(\hat{\beta}_1) \quad [E(\hat{\beta}_1) = \beta_1 \text{ proved above}]$$

$$= \frac{1}{n} \sum E(y_i) - \bar{x} \beta_1$$

$$= \frac{1}{n} \sum E(\beta_0 + \beta_1 x_i + \epsilon_i) - \bar{x} \beta_1$$



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$$= \frac{1}{n} \sum (\beta_0 + \beta_1 x_i + \epsilon_i) - \bar{x} \beta_1$$

$$= \frac{n\beta_0}{n} + \beta_1 \frac{\sum x_i}{n} + \frac{0}{n} - \bar{x} \beta_1$$

$E(x_i) = x_i$
cos it's
constant

$$= \beta_0 + \beta_1 \left(\frac{\sum x_i}{n} \right) + 0 - \bar{x} \beta_1$$

$$\text{but } \frac{\sum x_i}{n} = \bar{x}$$

$$\Rightarrow E(\hat{\beta}_0) = \beta_0 + \beta_1 (\bar{x}) - \beta_1 (\bar{x})$$

Hence Proved

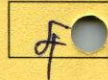
$$E(\hat{\beta}_0) = \beta_0$$

Therefore both
 $\hat{\beta}_1$ & $\hat{\beta}_0$
estimator

are unbiased
of β_1 & β_0 respectively

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(Q2)



Given

$$y_i = \beta_0 + \epsilon_i \quad \& \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$(i) \Rightarrow E(y_i) = E(\beta_0 + \epsilon_i) = \beta_0 + E(\epsilon_i) = \beta_0$$

$$\& \text{Var}(y_i) = \text{Var}(\beta_0 + \epsilon_i) = \text{Var}(\epsilon_i) = \sigma^2$$

Given

$$\hat{\beta}_{01} = k(2y_1) + \frac{y_2}{4} + ky_3$$

$$E(\hat{\beta}_{01}) = E(k(2y_1) + \frac{1}{4}y_2 + ky_3)$$

$$= 2kE(y_1) + \frac{1}{4}E(y_2) + kE(y_3)$$

$$= 2k(\beta_0) + \frac{1}{4}(\beta_0) + k(\beta_0)$$

$$= \beta_0 [2k + \frac{1}{4} + k]$$

for $\hat{\beta}_{01}$ to be unbiased estimator of β_0 :-

$$E(\hat{\beta}_{01}) = \beta_0$$

$$\Rightarrow 3k + \frac{1}{4} = 1 \Rightarrow 3k = \frac{3}{4} \Rightarrow k = \frac{1}{4}$$

$$\text{Ans (i)} \quad k = \frac{1}{4}$$

(ii) Now,

$$\hat{\beta}_{01} = \frac{y_1}{2} + \frac{y_2}{4} + \frac{y_3}{4} = \frac{2y_1 + y_2 + y_3}{4}$$

$$\& \hat{\beta}_{02} = \frac{y_1 + 2y_2 + 2y_3}{5} = \frac{y_1}{5} + \frac{2}{5}y_2 + \frac{2}{5}y_3$$



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I will choose the estimator which has min Variance.

Now,

$$\text{Var}[\hat{\beta}_{01}] = \text{Var}\left(\frac{y_1}{2} + \frac{y_2}{4} + \frac{y_3}{4}\right)$$

Since y_1, y_2 & y_3 are independent:

$$= \text{Var}\left(\frac{y_1}{2}\right) + \text{Var}\left(\frac{y_2}{4}\right) + \text{Var}\left(\frac{y_3}{4}\right)$$

We know:

$$\text{Var}(ax+b) = a^2 \text{Var}(x)$$

Hence

$$\begin{aligned} &= \frac{1}{4} \text{Var}(y_1) + \frac{1}{16} \text{Var}(y_2) + \frac{1}{16} \text{Var}(y_3) \\ &= \left[\frac{1}{4} + \frac{1}{16} + \frac{1}{16}\right](\sigma^2) \quad [\text{Var}(y_i) = \sigma^2] \\ &= 0.375 \sigma^2 \end{aligned}$$

$$\text{Var}[\hat{\beta}_{02}]$$

$$= \text{Var}\left[\frac{y_1}{5} + \frac{2}{5}y_2 + \frac{2}{5}y_3\right]$$

$$= \text{Var}\left(\frac{y_1}{5}\right) + \text{Var}\left(\frac{2}{5}y_2\right) + \text{Var}\left(\frac{2}{5}y_3\right)$$

$$= \frac{1}{25} \text{Var}(y_1) + \frac{4}{25} \text{Var}(y_2) + \frac{4}{25} \text{Var}(y_3)$$

$$= \left(\frac{1}{25} + \frac{4}{25} + \frac{4}{25}\right) \sigma^2 \quad [\text{Var}(y_i) = \sigma^2]$$

$$= \left(\frac{9}{25}\right) \sigma^2 = 0.36 \sigma^2$$

Since, $\text{Var}(\hat{\beta}_{02}) < \text{Var}(\hat{\beta}_{01})$

$\Rightarrow \hat{\beta}_{02}$ is better estimator of β_0 .

[Rationale: Efficient estimators have the minimum variance among all other available estimators]

(5)



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(Q3)

(i)

Model

Fitted Lin. Reg. Model

① $\ln(y \sim x_1)$

$$y = 2.20 + 0.6x_1$$

② $\ln(y \sim x_2)$

$$y = 2.5 + 0.5x_2$$

③ $\ln(y \sim x_1 + x_2)$

$$y = 2.2632 + 0.7895x_1 - 0.2105(x_2)$$

3

where

y = Quiz Marks (out of 10)

x_1 = "Hours of self study per week"

x_2 = "Hours spent watching lectures"

(ii) H_0 (Null Hypothesis):

$$E(y|x_1) = \beta_0$$

[or $\beta_1 = 0$]

"Hours of self study per week" & "Quiz Marks" have no linear relationship, hence model is not significant

H_a (Alternate Hypothesis) :

$$E(y|x) = \beta_0 + \beta_1 x_1 \quad [\text{or } \beta_1 \neq 0]$$

"Hours of Self study per week" &
"Quiz marks" have a linear relationship
and therefore model is significant.

From "Output 1"
we can see that p-value is 0.124
& we are given $\alpha = 0.05$

Since " $p < \alpha$ " is not true

Therefore we fail to Reject H_0

& conclude
 $\beta_1 = 0$

or
Model b/w y & x_1 is not significant.

Final conclusion should be according to the
question.



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(iii)

Model 1:

$$R^2 = 0.6$$

⇒ Around 60% of variation in y (Quiz marks) can be explained via ' x_1 ' (Hours of self study per week)

Since $R^2_{\text{(Model 1)}} > R^2_{\text{(Model 2)}}$

Model 1 will perform better as it captures & explains more variation in y (Quiz Marks).

(2)

Model 2:

$$R^2 = 0.4167$$

⇒ Around 41.67% of variation in y (Quiz Marks) can be explained by the ' x_2 ' (Hours spent watching lectures)

(Q4) Given $y_i = \beta_1 x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$

(i)

$$E(y_i) = E(\beta_1 x_i + \epsilon_i) = E(\beta_1 x_i) + E(\epsilon_i)$$

$$= \beta_1 x_i$$

$$\text{Var}(y_i) = \text{Var}(\beta_1 x_i + \epsilon_i) = \text{Var}(\epsilon_i)$$

$$= \sigma^2$$

Now,

$$RSS = \sum (\hat{y}_i - y_i)^2$$

but $\hat{y}_i = \beta_1 x_i$

$$\Rightarrow RSS = \sum (\beta_1 x_i - y_i)^2$$

To minimize RSS.

F.O.C

$$\frac{\partial RSS}{\partial \beta_1} = 2 \sum (\hat{\beta}_1 x_i - y_i)(x_i) = 0$$

$$\Rightarrow \sum [\hat{\beta}_1 x_i^2 - y_i x_i] = 0$$

$$\Rightarrow \hat{\beta}_1 \sum x_i^2 - \sum x_i y_i = 0$$

$$\Rightarrow \hat{\beta}_1 \sum x_i^2 = \sum x_i y_i$$

$$\Rightarrow \boxed{\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}}$$



(3)



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(ii)

Since

$$\hat{\beta}_1 = \frac{\sum (x_i y_i)}{\sum x_i^2}$$

$$\text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum x_i y_i}{\sum x_i^2}\right)$$

Since $\sum x_i^2 = \text{constant}$

$$= \frac{1}{(\sum x_i^2)^2} \text{Var}(\sum x_i y_i) = \frac{1}{(\sum x_i^2)^2} \cdot \sum \text{Var}(x_i y_i)$$

[Since y_i 's are independent $\Rightarrow \text{Var}(y_1 + y_2) = \text{Var}(y_1) + \text{Var}(y_2)$]

$$= \frac{1}{(\sum x_i^2)^2} \sum \text{Var}(x_i \cdot \beta_1 x_i) = \frac{\beta_1^2}{(\sum x_i^2)^2} \sum \text{Var}(x_i^2)$$

$$= \text{Var}\left(\frac{\sum x_i (\beta_1 x_i + \epsilon_i)}{\sum x_i^2}\right) = \text{Var}\left(\frac{\sum x_i^2 \beta_1}{\sum x_i^2} + \frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)$$

$$= \text{Var}\left(\beta_1 + \frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)$$

$$= \text{Var}\left(\beta_1 + \frac{\sum x_i \epsilon_i}{\sum x_i^2}\right)$$

$$= \cancel{\text{Var}(\beta_1)} + \text{Var}\left(\frac{\sum x_i \epsilon_i}{\sum x_i^2}\right) \rightarrow \frac{\sum x_i^2 \text{Var}(\epsilon_i)}{(\sum x_i^2)^2}$$

①

~~Var~~ Since error terms are not dependent on x_i (they are random)

$$= \left\{ \frac{\sum x_i}{\sum x_i^2} \right\}^2 \text{Var } \epsilon_i$$

$$\rightarrow \frac{\sigma^2 \sum x_i^2}{(\sum x_i^2)^2}$$

$$= \left(\frac{\sum x_i}{\sum x_i^2} \right)^2 \sigma^2$$

$$= \frac{\sigma^2}{\sum x_i^2}$$

$$\boxed{\text{Var}(\hat{\beta}_1) = \left(\frac{\sum x_i}{\sum x_i^2} \right)^2 \sigma^2}$$



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(iii)

Given

$$y_i = \beta_1 x_i + \epsilon_i \text{ \& } \epsilon_i \sim N(\mu, \sigma^2)$$

$$\Rightarrow E(y_i) = E(\beta_1 x_i) + E(\epsilon)$$
$$= \beta_1 x_i + \mu$$

0.5

$$\text{Bias}(\hat{\beta}_1) = E(\hat{\beta}_1) - \beta_1$$

$$= \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} - \beta_1$$

(5)

Linear Zero estimators:

There are the estimators which are linear in parametric space & also have Expected value 0.
 $\tilde{p}^T \tilde{\beta}$ is a linear zero estimator if it can be written as

$$\tilde{p}^T \tilde{\beta} = \mathcal{L}^T y$$

and

$$E(\tilde{p}^T \tilde{\beta}) = 0$$

Expected Value.

1.5



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